

DTSA 5720

Computing, Ethics, and Society 3 - Applications

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective	AI Engineer (60.4 / 100)	46.9 / 100
SPECIALIZATION Computing, Ethics, and Society		

47

CURRENT AVG SCORE
across 6 job roles

83

POTENTIAL SCORE
with recommended additions

+49

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Computing systems and technologies fundamentally impact the lives of most people in the world, including how we communicate, get information, socialize, and receive healthcare. This course is the third of a three course sequence that examines ethical issues in the design and implementation of computing systems and technologies, and reflects upon the broad implication of computing on our society. It covers medical applications, uses of robotics, autonomous vehicles, and the future of work.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
AI Engineer	60.4	98.4	+38.0	HIGH
Machine Learning Engineer	53.5	88.8	+35.4	HIGH
Data Scientist	43.3	74.7	+31.4	HIGH
ML Engineer	42.9	70.2	+27.4	MEDIUM
MLOps Engineer	41.2	77.5	+36.3	HIGH
Applied Scientist	40.2	88.9	+48.7	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

Business Communication

1 of 6 roles

business

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1

LOW

Optional: Business Communication — would broaden appeal to one additional role. Candidate terms: business.

GAP KEYWORDS BY ROLE

AI Engineer

60.4 → 98.4 (+38.0 pts)

perceptiondataengineeringlearninginfrastructuresoftwaremodelproblems

Machine Learning Engineer

53.5 → 88.8 (+35.4 pts)

datawebengineeringlearninganalytics_toolsexploratorybusiness

Data Scientist

43.3 → 74.7 (+31.4 pts)

dataanalyticssocialengineeringproblems

ML Engineer

42.9 → 70.2 (+27.4 pts)

media_discoverydatalearningmachinemodel_engineeringinfrastructure

MLOps Engineer

41.2 → 77.5 (+36.3 pts)

infrastructuredata_modellearningmachinemediain_engineering_tools

Applied Scientist

40.2 → 88.9 (+48.7 pts)

discovery_scientificresearch_problemsdata_learning_engineering_domains

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum–Industry Alignment project team.